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# Needleless Transcutaneous Neuromodulation Accelerates Postoperative Recovery Mediated via Autonomic and Immuno-Cytokine Mechanisms in Patients With Cholecystolithiasis

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**Background:** Postsurgical gastrointestinal disturbance is clinically characterized by the delayed passage of flatus and stool, delayed resumption of oral feeding, dyspepsia symptoms, and postsurgical pain. This study was designed 1) to evaluate the effects of needleless transcutaneous neuromodulation (TN) on postoperative recovery; 2) to investigate mechanisms of the TN involving autonomic functions in postoperative patients after removal of the gallbladder.

**Methods:** Sixty patients scheduled for laparoscopic cholecystectomy (LC) were randomized to TN ($n$ = 30) and sham-TN ($n$ = 30). TN was performed via acupoints ST36 and PC6 for 30 min twice daily from 24 hours before surgery to 72 hours after surgery. Sham-TN was performed using the same parameters at nonacupoints.

**Results:** 1) Compared to sham-TN, TN shortened time to first flatulence (38.9 $\pm$ 4.0 vs. 24.9 $\pm$ 2.4 hour, $p$ = 0.004) and time to defecation (63.1 $\pm$ 4.5 vs. 42.5 $\pm$ 3.1 hour, $p$ < 0.001). 2) Compared to sham-TN, TN increased the percentage of normal pacemaking activity (66.2 $\pm$ 2.2 vs. 73.8 $\pm$ 2.3%, $p$ = 0.018). 3) TN enhanced vagal activity. Compared to that 24 hours before surgery, surgery decreased vagal activity (HF) (0.41 $\pm$ 0.02 vs. 0.34 $\pm$ 0.02, $p$ = 0.043) 3 hours after the operation. Compared to sham-TN, TN increased HF (0.45 $\pm$ 0.02 vs. 0.52 $\pm$ 0.02, $p$ = 0.045) 72 hours after the operation. Further, HF was negatively correlated with time to defecation and serum norepinephrine. 4) Surgery increased serum IL-6 (1.1 $\pm$ 0.1 before surgery vs. 2.9 $\pm$ 0.7 pg/mL, $p$ = 0.041) 72 hours after the operation, which was reduced to baseline by TN (0.9 $\pm$ 0.1).

**Conclusions:** In conclusion, the proposed needleless TN accelerates postoperative recovery after LC, possibly mediated via the autonomic and immune-cytokine mechanisms. Needleless and self-administrable TN may be an easy-to-implement and low-cost complementary therapy for postoperative recovery.

**Keywords:** Inflammatory cytokines, laparoscopic cholecystectomy, postoperative recovery, transcutaneous electrical neuromodulation, vagal activity

**Conflict of Interest:** The authors declare no conflicts of interest related to this study.

## INTRODUCTION

Cholecystectomy for gallstone removal is common: >700, 000 cases/year and costs about $6.5 billion annually in the United States (1). The postsurgical gastrointestinal (GI) disturbance is a frequent complication of abdominal operation; its main symptoms include delayed passage of gas and stool, feeding intolerance, and dyspepsia symptoms (abdominal pain, nausea, abdominal distention, etc.) (2). Laparoscopic cholecystectomy (LC), despite small incision size, also results in postsurgical symptoms (3).

Early studies have suggested that open abdominal surgery induces sympathetic stress response (4,5) and inflammatory response which results in norepinephrine and interleukin 6 (IL-6) release, yielding a long-lasting phase of postoperative ileus (POI) (6). Sympathetic stress response was also reported in LC, attributed to anesthetics, small abdominal incisions and stretch of peritoneum due to carbon dioxide pneumoperitoneum and other nonspecific findings (6,7). In addition to sympathetic overactivity, manipulating and dissecting gallbladder and surrounding tissues increase proinflammatory cytokines IL-6 (6). It was well-known

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that autonomic neurological dysfunction plays an important role in the progression of impaired GI motility (8).

The conventional intervention to prevent postsurgical GI disturbance or restore GI motility includes Enhanced Recovery After Surgery programs, adequate pain control, early mobilization, epidural anesthesia (9), and drugs such as alvimopan, erythromycin among others (10). However, about 30–40% patients are generally unsatisfied with the conventional therapies (11). Therefore, finding novel and effective therapies to improve GI functional recovery has become an integral part of postoperative rehabilitation.

Neuromodulation has recently been shown effective in the management of GI motility disorders (12). In human studies, neuromodulation or electroacupuncture (EA) at ST36 (above of lateral sural cutaneous nerve, slender nerve, and deep peroneal nerve) has been demonstrated to accelerate postoperative GI function recovery (13). In animal studies, neuromodulation at ST36 has been reported to improve GI motility by enhancing vagal activity and inhibiting sympathetic activity (14,15). In addition to improving motility, neuromodulation at ST36 has also been shown to decrease inflammatory cytokine IL-6 in animals via vagal mechanisms (15–17). However, EA uses needles, and this limits widespread applications since they must be administrated by healthcare professionals.

A newly developed method of EA by replacing needles with surface electrodes is named TN. TN at PC6 (above of median nerve) has been shown to improve dyspepsia symptoms in patient with diabetes or undergoing chemotherapy (18). It has been reported that TN at PC6 and ST36 improves cold stress-induced impaired gastric accommodation (18,19) and chronic constipation (20). TN has been consistently shown to increase GI motility by activating the vagal nerve (8,21,22). However, it is unknown whether TN might have a therapeutic effect on postoperative recovery. We hypothesized that abdominal surgeries activate sympathetic never and proinflammatory cytokines, and therefore, impair GI motility, and that TN suppresses surgery-induced sympathetic overactivity and inflammation, and thus improves postoperative GI motility.

The aims of this study were to investigate the effects of TN on postoperative recovery and to study mechanisms of TN involving autonomic functions and proinflammatory cytokines in patients undergoing LC for gallstone removal.

## METHODS

### Study Participants

A prospective study was performed between June 2016 and December 2016 at the Yinzhou People's Hospital. The Ethics Committee of the hospital approved the study. Each patient signed an informed consent form before being recruited. Inclusion criteria were as follows: 1) age 18–65 years; 2) scheduled for LC surgery for gallstones, 3) willing to follow the treatment plan. Exclusion criteria: 1) diabetes mellitus, 2) using drugs which would affect GI motility, 3) changing LC to open surgery, 4) allergic to skin preparation, and 5) familiar with acupoints and their functions.

### Study Design

Patients with gallstones scheduled for LC were recruited and randomized into two groups: TN and sham-TN. Randomization: 1) We generated 56 random numbers by excel; 2) each enrolled subject got one random number; 3) random numbers were sorted by order, the first every other number were chosen as the treatment group and the rest as the control. The sample size calculation was performed as follows: according a previous study (13) and our predicted effect of TEA, the total sample size was 56 calculated by G power software packet. Software G power parameters were set as follows: 1) test family: *t* tests; 2) statistical test: means: difference between two independent means (two groups); 3) tail(s): two; effect size dz: 0.79; $\alpha$ err prob: 0.05; Power: 0.9; 4) primary outcome time to defecation was selected to calculate effect size: Effect size d: mean value of control group = 122 hours; standard deviation = 53 hours; mean value of treatment group = 86 hours; standard deviation = 36 hours. Other routine postoperative cares were basically the same between the two groups. Patients in each group underwent two sessions of TN or sham-TN daily from 24 hours before surgery to 72 hours after surgery. The dyspepsia symptoms score, visual analog scale (VAS) pain score, time of the first defecation, time of the first flatulence, time to resuming diet were noted. Electrocardiogram (ECG) and electrogastrogram (EGG) were recorded 24 hours before surgery, 3 hours, and 72 hours after surgery. A blood sample was taken at 7 am 24 hours before surgery, 24 hours, and 72 hours after surgery.

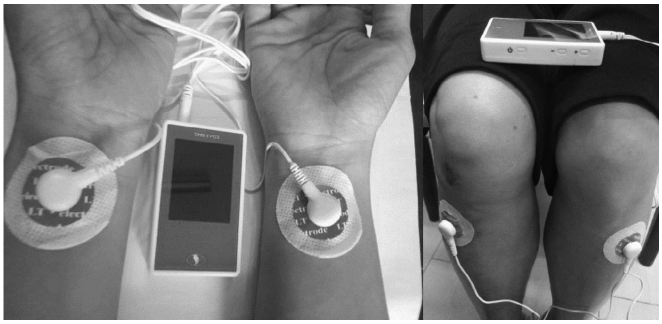


**Figure 1.** Method of transcutaneous electrical acustimulation. One electrode was placed on acupoint PC6 and another at the same place but the other hand, as well as the ST36 acupoint. A watch-size stimulator was connected to the electrodes for stimulation.

### Transcutaneous Neuromodulation and Sham-TN

TN was performed at two acupoints, ST36 and PC6 (Fig. 1) (20,21,23). The reasons we chose these points were 1) they were in the vicinity of peripheral nerves, ST36 is close to peroneal nerve, and PC6 coincides median nerve; 2) TN at these points were shown to improve GI motility and decrease sympathovagal balance (8,21,22). The location of ST36 is at the place of four-finger-breadth measuring down from the depression just lateral to the patellar tendon of the knee between the fibula and the tibia, one-finger-breadth measurement besides the tibia. The location of PC6 is at the junction of the distal one-sixth and the proximal five-sixth from the transverse crease of the wrist to the transverse cubital crease, between the tendons of palmaris longus and flexor carpi radialis. A pair of electrodes were placed at bilateral ST36 and another pair of electrodes were placed at bilateral PC6. A watch-size digital stimulator (SNM-FDC01, Ningbo Maida Medical Device, Inc., Ningbo, China) was used to deliver trains of pulses. The stimulation parameters for ST36 stimulator were set at 2 sec on and 3 sec off, pulse width of 0.5 msec, pulse frequency of 25 Hz, and amplitude of 2–10 mA (at the maximum level tolerated by the subject) based on previous studies (20,21,23); for PC6, stimulator was set at 0.1 sec on and 0.4 sec off, pulse width of 0.5 msec, pulse frequency of 100 Hz, and amplitude of 2–10 mA, which were shown to improve nausea and vomiting, and visceral pain (24,25). The bilateral stimulation TN treatment was given for half hour, twice daily (once on the day of surgery), lasting for 4 days.

Sham-TN stimulation was the same as the TN except that ST36 and PC6 were replaced by nonacupoints: not on any meridian, 15–20 cm away from PC6 (up to the elbow and lateral to elbow not on any meridian) and 10–15 cm down from and to the lateral side of ST36 were chosen as sham-PC6 and sham-ST36 (20,21,23). The investigators and patients were blinded to the nature of the therapy.

### Outcome Measurements

Main outcome measurements included time to the first defecation, time to first flatulence. Other measurements included time to resume diet, dyspepsia symptoms scores, and VAS scores. Dyspepsia symptoms were scored from 0 to 3 (none, mild, moderate, and severe) which included upper abdominal pain, upper abdominal discomfort, postprandial fullness, upper abdominal bloating, early satiety, nausea, vomiting, excessive belching, and heartburn. The VAS pain scores were used for assessing pain from 0 to 10; 0: painless; 1–3: mild and endurable pain; 4–6: moderate pain affecting sleep but still endurable; 7–10: severe pain, unbearable, affecting appetite and sleep.

### Recording and Analysis of Electrogastrogram

Gastric pace-making activity (slow waves) was assessed by a 4-channel portable EGG recorder (MEGG-04A, Ningbo Medkinetic, Inc., Ningbo, Zhejiang, China) (21). The subject was fasted for more than 6 hours and asked to lie supine. The abdominal skin where electrodes were to be placed was cleaned with special gel (Nuprep, Weaver, and Company, Aurora, USA). Then conductive gel (Ten20, Weaver and Company, Aurora, USA) was applied to reduce electrode–skin impedance. Then six electrocardiography electrodes were placed 1) xiphoid process, a reference electrode; 2) between the xiphoid process and the umbilicus, electrode 3; 3) 2–3 cm to the horizontal right of electrode 3, electrode 4; 4) level with reference electrode and 45° angle upper-left of electrode 3, electrode 1; 5) midpoint between electrode 1 and 3, electrode 2; 6) apical part, GND electrode (26,27). During the entire recording period, talking, reading, or sleeping was not allowed.

The EGG signals were analyzed using special spectral analysis software package (Ningbo MedKinetic, Inc., Ningbo, Zhejiang, China) (21). According to the tracing, motion artifacts were deleted and then running spectral analysis was performed. Clinically established EGG parameters (normal slow wave, bradygastria, and tachygastria) were derived from the EGG signals. Normal pace-making activity was defined as the dominant peak in the range of 2–4 cycles per minute (cpm); bradygastria and tachygastria were designated as the dominant peak in the range of 0.5–2 cpm and 4–9 cpm, respectively (28,29).

### Assessment of Autonomic Functions

A special amplifier with a recording frequency of 100 Hz (ECG-201, Ningbo Maida Medical Device, Inc., Ningbo, China) was used to record the ECG signal as follows: three locations where ECG electrodes were to be placed (apical, the second intercostal of sternal right edge, and the intersection of right rib arch and clavicle midline) were marked, gently rubbed with sandy gel and wiped clean with a gauze. Electrodes were placed over the treated skin after the skin is dry. The heart rate variability (HRV) signal was derived using a custom-made software by identifying the R peak and calculating the RR interval from the ECG. The power spectrum analysis of HRV signal is used to calculate the power of different frequency bands. The power in the low-frequency (LF) band (0.04–0.15 Hz), LF stands mainly sympathetic activity, whereas, the power in the high-frequency (HF) band (0.15–0.50 Hz), HF, reflects purely parasympathetic or vagal activity. In this study, the standardized values are used: that is, LF is LF/(HF + LF), and HF is HF/(HF + LF) (25,30).

### Blood Draw and Assay

A blood sample was taken at 7 am 24 hours before surgery, 24 hours, and 72 hours after surgery. The 5 mL blood was stored in the procoagulant tube and then centrifuged by 3000 rpm for 5 min. The serum was divided into three portions, each of which was about 0.5 mL, and was stored at −80°C for 6 months. Human ELISA kits (Roche Company, American, Cat. No. 0734–2014, Shanghai Shiyan Biological Technology Co., Ltd, China, Lot. No. 201704) were used to analyze IL-6 and norepinephrine (NE), respectively.

### Statistical Analysis

All data are expressed as mean ± standard error (SE). $p < 0.05$ was considered statistical significance. Student's *t*-test was performed to assess the difference in the measurement (time to defecation, time to flatulence, time to resume diet) between sham-TN and TN group. Between the groups, repeated measurement analysis of variance was used to compare the difference in dyspepsia symptoms scores, VAS pain scores, HF (vagal activity), and LF/HF (sympathovagal ratio) at 24 hours before surgery, 3 hours, and 72 hours after surgery and serum NE and IL-6 at 24 hours before surgery, 24 hours, and 72 hours after surgery. Data were analyzed by statistical software SPSS 19.0.

## RESULTS

### Characteristics of Patients

A total of 60 patients were recruited and randomized for sham-TN (30 patients) or TN (30 patients). Demographic data and surgical details are shown in Table 1. The gender, age or body mass index showed no difference between the two groups (all $p > 0.05$ for each). No difference was noted either in surgical procedures between the two groups.

**Table 1.** Characteristics of Patients and Surgical Details.

| Observation index | Sham-TN | TN |
|---|---|---|
| Patient (*n*) | 30 | 30 |
| Acute biliary colic/cholecystitis/ asymptomatic gallstones (*n*) | 6/16/8 | 7/15/8 |
| Male (*n* [%]) | 19 [63.3] | 17 [56.6] |
| Age (mean ± SE) | 56.5 ± 2.4 | 57.1 ± 2.0 |
| BMI (mean ± SE, kg/m$^2$) | 23.7 ± 0.7 | 25.0 ± 0.5 |
| Surgical time (mean ± SE, min) | 55.0 ± 5.7 | 53.3 ± 4.9 |

BMI, body mass index.



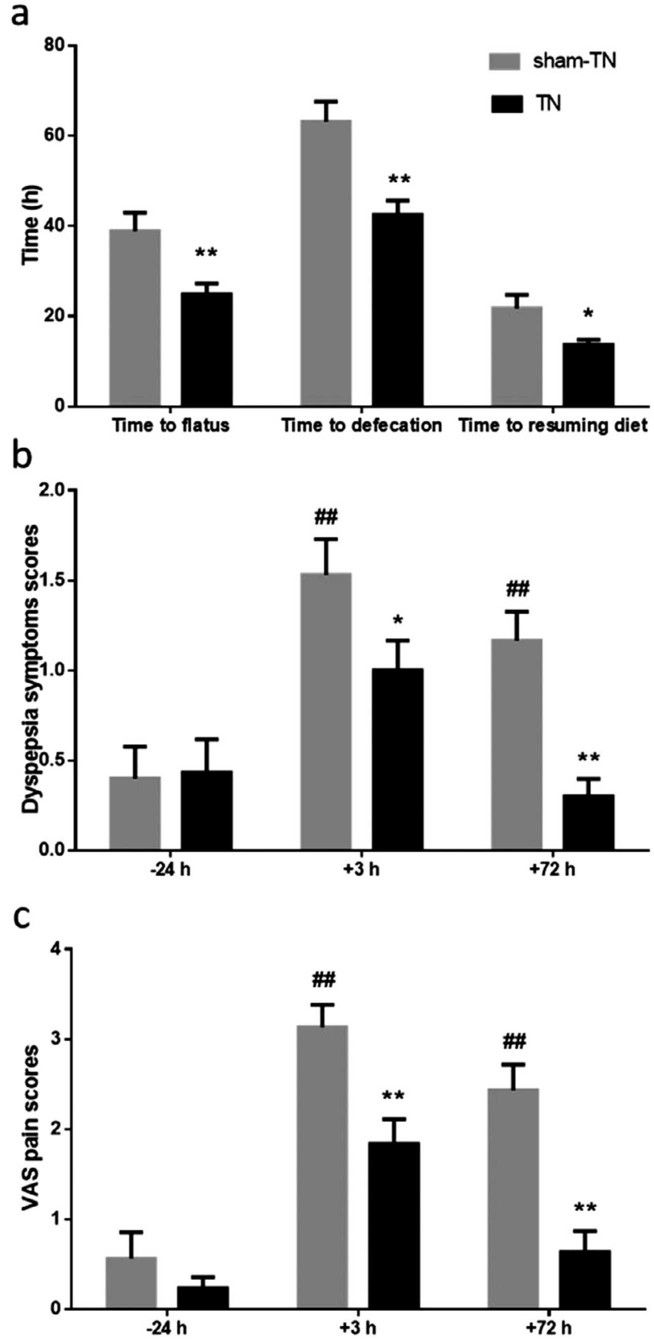



**Figure 2.** Effects of TN on postoperative clinical symptoms. TN shortened time to defecation and time to flatus, as well as time to resuming diet (a). Gastrointestinal symptoms (b) and VAS pain scores (c) in patients 24 hours before laparoscopic cholecystectomy, and 3 hours, 72 hours after the operation. (vs. sham-TN, *: $p < 0.05$; **: $p < 0.01$; vs. before surgery, #: $p < 0.05$; ##: $p < 0.01$).

## TN Improved Postoperative Recovery

TN shortened the postsurgical recovery time. Compared to sham-TN, TN shortened time to first flatulence (38.9 ± 4.0 vs. 24.9 ± 2.4 hour, $p = 0.004$), time to first defecation (63.1 ± 4.5 vs. 42.5 ± 3.1 hour, $p < 0.001$) and time to resuming diet (21.7 ± 3 vs. 13.7 ± 1 hour, $p = 0.014$) (Fig. 2a).

Compared to that 24 hours before surgery, surgery increased the dyspepsia symptom score (0.40 ± 0.18 vs. 1.53 ± 0.20, $p < 0.001$) 3 hour and (0.40 ± 0.18 vs. 1.17 ± 0.16, $p = 0.003$) 72 hours after the operation. In comparison with sham-TN, TN decreased the dyspepsia symptoms score (1.53 ± 0.20 vs. 1.00 ± 0.17, $p = 0.042$) 3 hours and (1.17 ± 0.16 vs. 0.30 ± 0.10, $p < 0.001$) 72 hours after the operation (Fig. 2b).

Compared to that 24 hours before surgery, surgery increased the VAS pain score (0.50 ± 0.23 vs. 3.13 ± 0.25, $p < 0.001$) 3 hours and (0.50 ± 0.23 vs. 2.43 ± 0.29, $p < 0.001$) 72 hours after the operation. Compared with sham-TN, TN decreased the VAS pain score (3.13 ± 0.25 vs. 1.83 ± 0.30, $p = 0.001$) 3 hours and (2.43 ± 0.29 vs. 0.63 ± 0.24, $p < 0.001$) 72 hours after the operation (Fig. 2c). There was no significant difference in postoperative hospital stay between sham-TN and TN group (3.46 ± 0.28 vs. 3.42 ± 0.25 days, $p = 0.924$).

## TN Improved Gastric Motility (Pace-Making Activity)

Compared to 24 hours before surgery, surgery decreased the percentage of normal pace-making activity (68.8 ± 1.5 vs. 53.1 ± 1.9%, $p < 0.001$) 3 hours after the operation. Compared to sham-TN, TN increased the percentage of normal pace-making activity (53.1 ± 1.9 vs. 59.8 ± 2.2%, $p = 0.025$) 3 hours, and (66.2 ± 2.2 vs. 73.8 ± 2.3%, $p = 0.018$) 72 hours after the operation (Fig. 3a).

Compared to 24 hours before surgery, surgery increased the percentage of bradygastria (12.9 ± 0.8 vs. 21 ± 1.1%, $p < 0.001$) 3 hours after the operation. Compared to sham-TN, TN decreased the percentage of bradygastria (21 ± 1.1 vs. 16.8 ± 1.3%, $p = 0.02$) 3 hours, and (13.4 ± 1.2 vs. 8.7 ± 0.9%, $p = 0.002$) 72 hours after the operation; compared to 24 hours before surgery, TN decreased the percentage of bradygastria (13.2 ± 1.1 vs. 8.7 ± 0.9%, $p = 0.006$) 72 hours after the operation (Fig. 3b). Compared to 24 hours before surgery, surgery increased the percentage of tachygastria (11.9 ± 0.8 vs. 19.7 ± 1.1%, $p < 0.001$) in patients 3 hours, and (11.9 ± 0.8 vs. 15.3 ± 1.4%, $p = 0.039$) 72 hours after the operation; TN reduced the percentage of tachygastria to a level comparable to that before surgery (Fig. 3c).

## Mechanisms of TN Involving IL-6

Compared to 24 hours before surgery, surgery increased serum IL-6 (1.1 ± 0.1 vs. 3.0 ± 0.9 pg/mL, $p = 0.026$) 24 hours and (1.1 ± 0.1 vs. 2.9 ± 0.7 pg/mL, $p = 0.041$) 72 hours after the operation. Compared to sham-TN, TN decreased serum IL-6 (2.9 ± 0.7 vs. 0.9 ± 0.1 pg/mL, $p = 0.009$) 72 hours after the operation (Fig. 4).

## Mechanisms of TN Involving Autonomic Functions

TN enhanced vagal activity. Compared to 24 hours before surgery, surgery decreased vagal activity (HF) (0.41 ± 0.02 vs. 0.34 ± 0.02, $p = 0.043$) 3 hours after the operation. Compared to sham-TN, TN increased HF (0.45 ± 0.02 vs. 0.52 ± 0.02, $p = 0.045$) 72 hours after the operation. Interestingly, compared to 24 hours before surgery, TN increased HF (0.43 ± 0.03 vs. 0.52 ± 0.02, $p = 0.016$) 72 hours after the operation (Fig. 5a). Similarly, the sympathovagal ratio (LF/HF) was increased by surgery ($p < 0.05$ vs. before surgery) and decreased by TN to a level comparable to the baseline before surgery ($p < 0.05$ vs. before surgery; Fig. 5b).

Compared to 24 hours before surgery, surgery increased serum NE (270.1 ± 18.3 vs. 544.7 ± 88.4 pg/mL, $p = 0.013$) 24 hours, and (270.1 ± 18.3 vs. 605.5 ± 65.8 pg/mL, $p = 0.003$) 72 hours after the operation. Compared to sham-TN, TN decreased serum NE by (605.5 ± 96.8 vs. 347.6 ± 61.3 pg/mL, $p = 0.028$)72 hours after the operation (Fig. 5c). The HF was negatively correlated with NE in

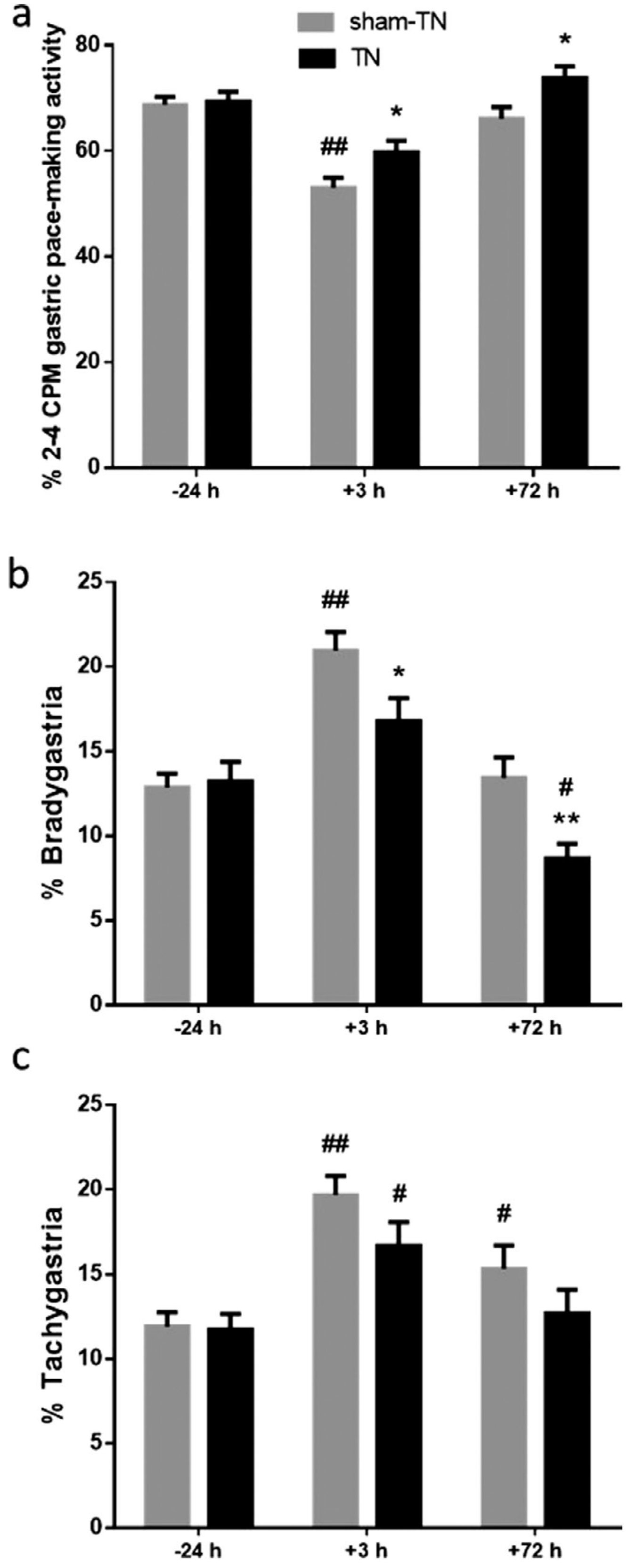



**Figure 3.** Gastric slow wave in patients 24 hours before laparoscopic cholecystectomy, and 3 hours, 72 hours after the operation. Surgery decreased %N pace-making activity and TN suppressed the decrease (a). Surgery increased % Bradygastria (b) and % Tachygastria (c) and TN improved it (vs. sham-TN, *: $p < 0.05$; **: $p < 0.01$; vs. before surgery, #: $p < 0.05$; ##: $p < 0.01$).

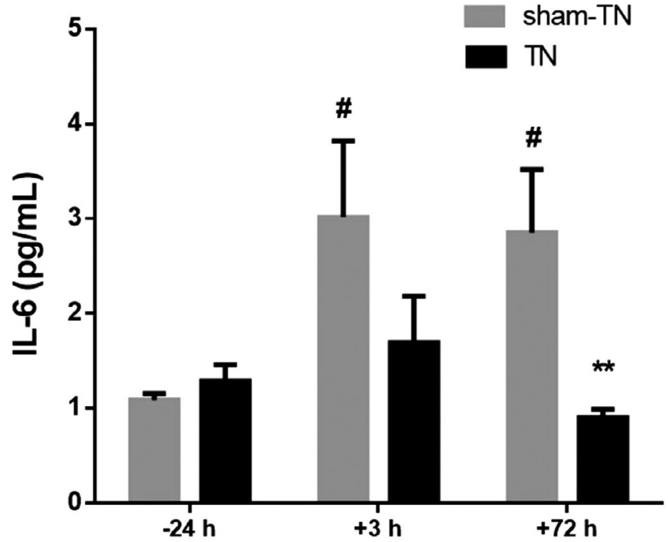



**Figure 4.** Effects of TN on inflammation. TN improved surgery induced serum IL-6 increase (vs. sham-TN, *: $p < 0.05$; **: $p < 0.01$; vs. before surgery, #: $p < 0.05$)

all patients ($r = -0.449$, $p < 0.01$), suggesting an agreement between these two different measures of the autonomic function.

In addition to the direct effect of TN on the vagal activity assessed by the HF, the HF was also found to be associated with a number of clinical outcome measures as shown below:

Correlation of Autonomic Functions With Clinic Symptoms

The HF 3 hours after surgery was negatively correlated with time to defecation ($r = -0.491$, $p < 0.01$) (Fig. 6).

Correlation of HF With Percentage of Normal Slow Waves

The HF was positively correlated with the percentage of normal pace-making activity ($r = 0.260$, $p < 0.01$).

Correlation of NE With IL6

The NE was positively correlated with IL-6 ($r = 0.720$, $p < 0.001$) in all patients.

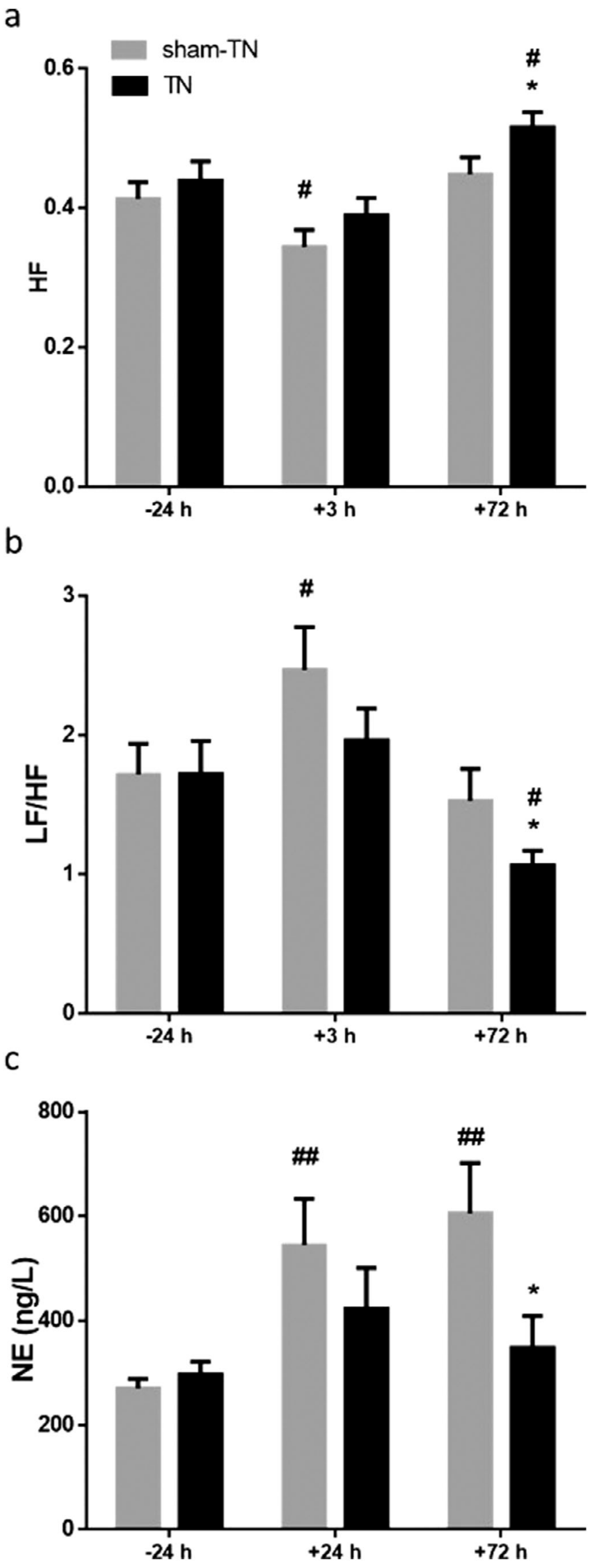
## DISCUSSION

In the current study, we found that: 1) TN improved major postoperative symptoms, including a reduction in time for the first defecation, time for the first flatulence, time to resume diet and dyspepsia symptoms as well as pain. 2) TN improved gastric motility (pace-making activity) impaired by surgery. 3) TN enhanced vagal activity and suppressed sympathetic overactivity assessed by the spectral analysis of HRV as well as serum NE. 4) The vagal activity, HF, assessed from the spectral analysis of HRV was negatively correlated with time to defecation and serum NE. Whereas, HF was positively correlated with the percentage of normal pacemaking activity. 4) TN significantly suppressed surgery-induced increase in serum IL-6 and IL-6 was found to be positively correlated with serum NE.

While the ameliorating effects of neuromodulation with needle-based EA on postoperative recovery had been reported (13), it was unknown whether needleless TN had similar effects. The present study was therefore designed to investigate the effects of needleless TN on postoperative recovery and its mechanisms





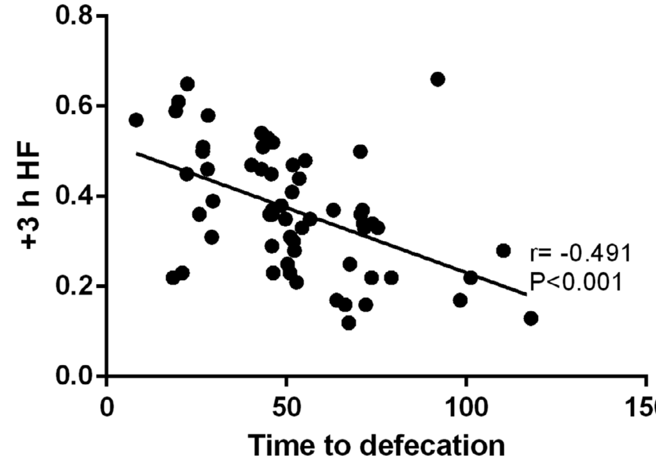




**Figure 6.** Correlation between vagal activity and clinic symptoms. HF 3 hours.

involving autonomic functions and proinflammatory cytokines. In this study, we used two special sets of parameters that were known to improve motility (20,21,23) and dyspepsia symptoms (24,25), respectively. The needleless method of TN can be self-administrated or administrated by nonprofessionals and is expected to be well received by patients. It can be easily incorporated into Enhanced Recovery After Surgery programs. To minimize placebo effects or biases, the study was designed as prospective, placebo-controlled and randomized, and the sham-TN was performed with the same electrical stimulation but at nonacupoints (8,19,21).

Postsurgical GI disturbance is common in patients after abdominal surgery (31), whereas, the pharmacological treatment was reported to show limited effects. In 2008, a Cochrane review included 39 randomized controlled trials reported that except Alvimopan, 14 systemic prokinetic drugs showed no beneficial effects on the first defecation time or recovery time in patients after surgeries; however, the authors suggested that Alvimopan should still be considered experimental due to methodological biases in the randomized clinical trials. (32). In 2012, a meta-analysis reported that Alvimopan was beneficial in shortening open abdominal surgical recovery but pointed out that the beneficial effects on laparoscopic surgeries could not be confirmed (33). In 2017, prucalopride was reported to improve postsurgical recovery and reduce inflammatory response (34); however, Smart and Malik (35) pointed out that multicenter randomized controlled trials were needed to make sure which kind of patients would gain benefits from this medicine.

Neuromodulation using needle-based electrical stimulation at acupoints has been reported to improve gastric motility and dyspepsia symptoms in patients with GI motility disorders, such as functional esophageal disorders, nausea and vomiting, functional dyspepsia, and constipation (36,37). In the current study, the needleless TN with parameters previously shown to improve gastric motility substantially reduced major postoperative symptoms. Reduced time to first bowel movement was also reported in a previous study in patients after laparoscopic colectomy for

**Figure 5.** Effects of TN on autonomic function in patients 24 hours before laparoscopic cholecystectomy, and 3 hours, 72 hours after the operation. Surgery decreased HF and increased LF/HF; TN increased HF (a) and decreased LF/HF (b). TN decreased serum NE increased by surgery (c) (vs. sham-TN, *: $p < 0.05$; **: $p < 0.01$; vs. before surgery, #: $p < 0.05$; ##: $p < 0.01$).



removal of colorectal cancers; however, the treatment was needle-based EA (13). Previously, similar TN but with different parameters was reported to improve postoperative pain in patients after low intraabdominal surgery (38) and after total hip arthroplasty (39). To the best of our knowledge, the present study was the first comprehensive trial using the needleless TN to treat postsurgical GI disturbances and its possible mechanisms in patients after LC.

In addition to improvement in postsurgical symptoms, the proposed TN also improved pace-making activity impaired by the surgery. The cutaneous EGG method used in this study has been proven to be a reliable and accurate noninvasive technique to record gastric pace-making activity (40) which is positively correlated with gastric motility (41). A number of previous studies reported that EA or TN using appropriate parameters improved pace-making activity and GI motility in animals (14,42–44) and patients (20–22). GI motility was not assessed in this study on account of invasiveness of the actual motility procedure and severity of the disease and surgical procedures in these patients; instead, we used the assessment of pace-making activity as a surrogate of gastric motility. In addition, it was previously indicated that (45) the combination of time to defecation and tolerance of solid food was the best indication of GI motility recovery. In our study, TN shortened time to defecation and resuming diet as well as improving pace-making activity. Accordingly, we speculated that acceleration of postsurgical recovery was attributed to TN-induced improvement in GI motility.

Our findings seemed to suggest that both surgery-induced postoperative symptoms and TN-induced improvement in surgical recovery were mediated via the autonomic mechanisms. In the current study, we discovered that the TN enhanced vagal activity assessed by the spectral analysis of HRV that was suppressed by surgery and suppressed sympathetic activity assessed by serum NE that was activated by surgery, resulting in a more balanced sympathovagal ratio. Moreover, the vagal activity was found to be negatively correlated with the time to defecation but positively correlated with the percentage of normal pace-making activity. Similar autonomic mechanisms were also noted in previous EA and TN studies in other conditions. EA at ST36 and PC6 with similar parameters has been consistently shown to increase vagal tone and decrease sympathovagal balance in animals with impaired gastric motility (14,15,25,46). The needleless TN has also been noted to be able to enhance vagal activity and decrease sympathovagal balance in patients with functional GI disorders (8,20,21,27). In addition to enhancing vagal activity, we also found TN decreased the serum level of NE that is considered as an indicator of sympathetic neural activity (47). Further, we found that the vagal activity (HF) was negatively correlated with serum NE. These interesting findings that had never been reported previously suggested again the autonomic mechanisms involved in the improvement of postoperative symptoms with the TN. We further speculated that the proposed TN suppressed sympathetic activity and enhancing vagal activity, and resulted in improvement in GI motility, leading to accelerated postoperative recovery.

Our results also suggested the involvement of immune-cytokine mechanisms with the TN. In the present study, the TN substantially decreased serum IL-6 72 hours after the operation in comparison with sham-TN. In previous studies (15,16), needle-based EA was noted to decrease IL-6 by enhancing vagal activity. Vagal activity was reported to be able to attenuate intestinal inflammation by reducing the secretion of IL-6 via activating $\alpha 7$ nicotinic acetylcholine receptors in macrophages in a mouse model of intestinal manipulation (48,49). After open abdominal surgery, macrophages are rapidly activated, resulting in release of inflammatory cytokines, such as IL-6 (47,50); further IL-6 activates inhibitory neural pathway, resulting in GI disturbances (51). A similar inflammatory response was also noted after LC (6,52), due to manipulating and dissecting the gallbladder and surrounding tissues. Inhibition of macrophage has been shown to prevent intestinal inflammation-induced POI (47,53,54).

The following neural pathway is speculated for the anti-inflammatory effect of TN in this study: first, TN activated the somatic nerve and sent a signal to the brain via the spinal afferent nerve as reported in a separate study in our lab (55). Then the brain processed the signal and provided an enhanced vagal efferent activity (56) Finally, acetylcholine was released (55) and acted on alpha 7 subunit of the nicotinic receptors in GI macrophages to balance the release of inflammatory cytokines (57).

Limitations of the present study included: 1) a longer duration of hospital stay was noted in the present study compared to that in west countries. This was attributed to social and culture differences, as well as hospital costs. In China, surgery, even LC, is considered a severe event and the patient takes it extremely seriously and both the patient and the healthcare professional use extra caution and measures to ensure a complete postoperative recovery before discharge. In addition, the cost of extended hospitalization is much lower than that in western countries. 2) The time taken for postsurgical recovery following LC seemed a little slow in this cohort. The exact reasons were unclear; however, the conservative attitude of the patient regarding early postsurgical oral feeding might be one of unknown factors. Traditionally in China, patients do not want to eat early after surgery and believe it is safer to resume food later. On the other hand, however, it has become clear recently that early oral feeding would facilitate postsurgical recovery as well as recovery of GI motility. 3) This was a one single-center study and no motility hormones were measured.

In conclusion, the proposed needleless TN accelerates postoperative recovery after LC possibly mediated via the autonomic and immune-cytokine mechanisms. Needleless and self-administrable TN may be considered as an easy-to-implement and low-cost complementary therapy for postoperative recovery and/or incorporated into the Enhanced Recovery After Surgery Programs.

## Authorship Statements

Drs. Liang Zhu, Bo Zhang, and Jiande D. Z. Chen designed the study. Drs. Bo Zhang, Kelei Zhu, Pingping Hu, and Feng Xu conducted the study including patient recruitment, data collection, and data analysis. Drs. Liang Zhu, Jiande D. Z. Chen, and Bo Zhang completed the manuscript writing and edition. All authors approved the final manuscript.

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## COMMENT

Neuromodulation to aid GI recovery after surgery is an under-investigated research area. The authors of this paper are to be commended for undertaking a randomized controlled trial design including a sham stimulation arm. Outcomes appear to be favorable for the selected stimulation strategy, and it would be interesting to see if these benefits translate to enhanced recovery in more major gastrointestinal procedures in future. Mechanisms were also evaluated, although the exact reasons for the enhanced outcomes remain somewhat unexplained.

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Comments not included in the Early View version of this paper.





  